

In [2]: `import numpy as np`

```
alpha = np.pi/3
beta = np.pi/4
delta = np.pi/4
gamma = 3*np.pi/2

def exp(theta):
    return np.cos(theta)+1j*np.sin(theta)

A1 = np.array([[exp(-beta/2), 0],
               [0, exp(beta/2)]])

A2 = np.array([[np.cos(gamma/2), -np.sin(gamma/2)],
               [np.sin(gamma/2), np.cos(gamma/2)]])

A3 = np.array([[exp(-delta/2), 0],
               [0, exp(delta/2)]])

global_phase = exp(alpha)
U = global_phase*A1@A2@A3

print("Matrix U is: ", U)
```

```
Matrix U is: [[-0.6830127 -0.1830127j -0.35355339-0.61237244j]
 [ 0.35355339+0.61237244j  0.1830127 -0.6830127j ]]
```

In []: